



FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF INFORMATION AND COMMUNICATION ENGINEERING

Second Semester Examination

Course Title: Operating System (3 Units)

Instruction: Answer 2 questions from each section.

2023/2024 Academic Session

Course Code: ICT 302

Time: 2 Hours

SECTION A

1. (a) With the aid of a diagram, explain operating system services. (5 Marks)
- (b) Using pictorial representation, describe six (6) processes represented in the operating system by a process control block (PCB). (5 Marks)
- (c) List the advantages of peer-to-peer systems over client-server systems. (3 Marks)
- (d) Explain Hybrid-Kernel Structure. (2 Marks)

2. (a) Explain the difference between symmetric and asymmetric multiprocessing. (2 Marks)
- (b) Explain the four basic module support components for Linux. (2 Marks)
- (c) Discuss the following: (5 Marks)
 - (i) Driver Registration and Conflict Resolution
 - (ii) Kernel and Completely Fair Scheduler (CFS)
- (d) With the aid of diagrams, explain the following: (6 Marks)
 - (i) The Linux physical memory zones
 - (ii) Memory layout for ELF programs

3. (a) In the context of an Operating System, differentiate between a thread and a process. (4 Marks)
- (b) Differentiate between Pre-emptive and Non pre-emptive Scheduling. (3 Marks)
- (c) List and briefly explain the criteria used in the selection of a CPU Scheduling Algorithm. (3 Marks)

(d) List six scheduling algorithm and briefly explain any 2 of them.

(5 Marks)

SECTION B

4. (a) Describe the concept of Multithreading in Operating Systems. (3 marks)

(b) Outline its benefits and provide at least two examples. (7 marks)

(c) Explain the necessity of Memory Management. (5 Marks)

5. (a) Identify and explain the two types of process execution in an Operating System.

(4 Marks)

(b) With the aid of a diagram, briefly explain process creation. (5 Marks)

(c) Differentiate between process management and file management. (6 Marks)

6. (a) Differentiate between LINUX and UNIX. (5 Marks)

(b) List four key features of Macintosh Operating Systems. (4 Marks)

(c) With the aid of a diagram, briefly explain the architecture of the Android operating system. (6 Marks)